BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 245 · Software development methods: evaluation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Independent tasks take 3 and 6 min; final integration 2 min. What is ideal parallel duration?

Q2. Name one XP practice intended to find code errors early.

Q3. Why would a user evaluate a prototype?

Q4. Simplify F=(¬A ∧ ¬B) ∨ (¬A ∧ B). State the law and final answer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

evaluation — Judgement supported by evidence and limitations.

risk-driven — Development priorities guided by significant uncertainties.

iterative — Revisits improvements in cycles.

stakeholder — Person or organisation affected by a project.

New learning

Avoid a universal answer — Method suitability depends on risk, requirements, regulation, user access and timescale.

Support with scenario evidence — A high-risk system may justify Spiral's explicit risk analysis despite extra overhead.

Include a drawback — Evaluate cost, documentation, flexibility and technical quality rather than merely naming a method.

Worked example

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

1. Predict the method — Recommend Spiral because technical risks can be evaluated at each iteration.

2. Check and explain — Explain the disadvantage: risk analysis adds project management and cost.

We Do · Practise with guidance

A small charity wants a booking page within a month and volunteers can review mock-ups daily.

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Rapid prototypes + daily feedback
→ RAD approach (with time for testing)
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Predict the method

RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.

Check and explain

Qualify the recommendation: frequent prototyping must not replace security and validation testing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give one defining feature each of Waterfall, Spiral, RAD and XP.

Question 2

Choose a development method for a regulated safety-critical control application with uncertain technology. Give three reasons and one drawback.

Question 3

Evaluate Agile compared with Waterfall for a government reporting system where requirements are partly fixed but user interface needs may change.

Exit Ticket · Answer before checking

The best development methodology is the same for every software project.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. max(3,6)+2=8 minutes.

Q2. Test-driven development or continuous integration.

Q3. To identify missing needs, usability issues or changes before more costly development.

Q4. Factor ¬A: F=¬A∧(¬B∨B)=¬A∧1=¬A. This also forms a pair of adjacent 1s on a K-map.

You Do pages · Independent questions

Q1. [4] Waterfall sequential stages [1]; Spiral risk-driven iterations [1]; RAD rapid prototyping with users [1]; XP practices such as TDD/pair programming [1].

Q2. [6] Spiral [1]; explicit risk analysis [1]; repeated validation [1]; manage technical uncertainty [1]; formal review [1]; greater cost/overhead [1].

Q3. [6] Agile permits feedback and changed UI requirements [2]; Waterfall offers well-defined approval/documentation milestones [2]; propose phased/hybrid governance or justify one method [1]; discuss risk of late change or fixed-scope rigidity [1].

Exit Ticket · Model answer

False. Project constraints determine the appropriate approach.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.